### 1. Competency #3: Implement a given strategy to determine a mathematical solution

- Highly relevant. The question implicitly provides a standard strategy: average speed = distance ÷ time.
- Students must apply this formula to each segment:
  - First segment: 4 km / 10 min = 0.4 km/min
  - Second segment: 2 km / 5 min = 0.4 km/min
- This leads to the conclusion that the speeds are equal, so option B is correct.
- This directly demonstrates the ability to "implement a known strategy to find a solution."

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### 2. Competency #7: Manipulate numbers, graphical and statistical data and information, algebraic expressions and equations, and geometric representations

- Highly relevant. The problem involves basic numerical manipulation—dividing distance by time.
- Although there are no graphs, algebraic symbols, or geometric elements, the core task is operating on numeric data.

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### 3. Competency #1: Perform a simple calculation

- Moderately relevant. Calculating 4 ÷ 10 and 2 ÷ 5 involves straightforward arithmetic.
- However, this competency is foundational and serves as a component of higher-level skills like #3 and #7.

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### 4. Competency #8: Articulate a solution, showing and/or summarising and presenting intermediate mathematical results

- Mildly relevant. Since the question is multiple-choice, students aren't required to explain their reasoning.
- However, if used in a formative assessment where justification is expected, this competency would become more pertinent.

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### 5. Competency #13: Use an understanding of the context to guide or expedite the mathematical solving process

- Weakly relevant. The real-world context (cycling, speed) helps ground the concept of "average speed," but it doesn't significantly influence the solution strategy.
- The scenario mainly provides data rather than shaping mathematical reasoning.

### 1. 能力 #3：Implement a given strategy to determine a mathematical solution

**（实施给定策略以确定数学解）**

- 高度相关。题目隐含了一个标准策略：平均速度 = 距离 ÷ 时间。
- 学生需应用这一公式分别计算两个阶段的速度：
  - 第一阶段：4 km / 10 min = 0.4 km/min
  - 第二阶段：2 km / 5 min = 0.4 km/min
- 由此得出速度相同，选择 B。
- 这直接体现了“实施已知策略求解”的能力。

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### 2. 能力 #7：Manipulate numbers, graphical and statistical data and information, algebraic expressions and equations, and geometric representations

**（操作数字、数据、代数表达式等）**

- 高度相关。本题涉及对数值（距离、时间）进行除法运算，属于基本的数值操作。
- 虽无图表或代数符号，但“操纵数字信息”是核心步骤。

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### 3. 能力 #1：Perform a simple calculation

**（执行简单计算）**

- 中等相关。计算 4÷10 和 2÷5 属于简单算术。
- 但此能力较基础，是更高阶能力（如 #3 和 #7）的组成部分。

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### 4. 能力 #8：Articulate a solution, showing and/or summarising and presenting intermediate mathematical results

**（清晰表述解法，展示中间结果）**

- 中等偏弱相关。虽然题目是选择题，不要求书面解释，但若在教学或评估中要求学生说明理由，则涉及此项。
- 在形成性评价中，此项可能被强调；但在本题形式下，相关性有限。

### 5. 能力 #13：Use an understanding of the context to guide or expedite the mathematical solving process

**（利用情境理解来指导或加速解题）**

- 弱相关。本题情境（骑自行车、速度计）有助于理解“平均速度”的现实意义，但解题过程并不依赖深层情境推理。
- 情境主要用于提供数据，而非影响策略选择。